

THE SEA MONSTER

Cetus



SIZE RANKING 4

BRIGHTEST STAR

Diphda (β) 2.0

GENITIVE Ceti

ABBREVIATION Cet

HIGHEST IN SKY AT 10 P.M.
October–December

FULLY VISIBLE

65°N–79°S



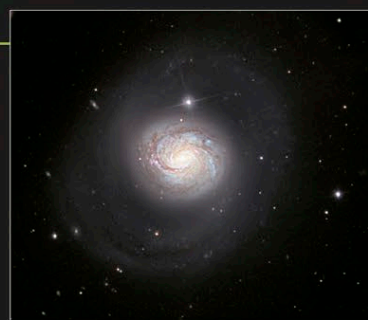
Cetus is represented on old star charts as an unlikely-looking, almost comical, hybrid sea monster, although the figure is also sometimes referred to as a whale. It is one of the original 48 Greek constellations listed by Ptolemy in his *Almagest*. It is a large but not very obvious constellation found in the equatorial region of the sky, and it lies south of the zodiacal constellations Pisces and Aries. Cetus is home to the celebrated variable star, Mira (ο), as well as a peculiar spiral galaxy, M77.

SPECIFIC FEATURES

Menkar (α) is the second-brightest star in the constellation. It forms part of the loop of stars that mark the sea monster's head, and it has a wide and unrelated 6th-magnitude companion that is visible through binoculars. Positioned near the neck of the sea monster is Gamma (γ) Ceti, a close double star

that is more challenging to divide than Menkar. High magnification on a telescope is required to see the two component stars, of 4th and 7th magnitudes.

Mira (ο) is the prototype of a common type of red giant that pulsates in size over months or years. Mira can reach magnitude 2 at its brightest—although magnitude 3 is more usual—while at its faintest it drops to magnitude 10. Hence, depending on how much it has swollen or contracted within its 11-month cycle, Mira can be either a naked-eye star or one that is visible only with a telescope.



Tau (τ) Ceti is 11.9 light-years away. Its temperature and brightness make it the most Sun-like of all Earth's nearby stars. Tau is, however, surrounded by a swarm of asteroids and comets, which would subject any local planets to

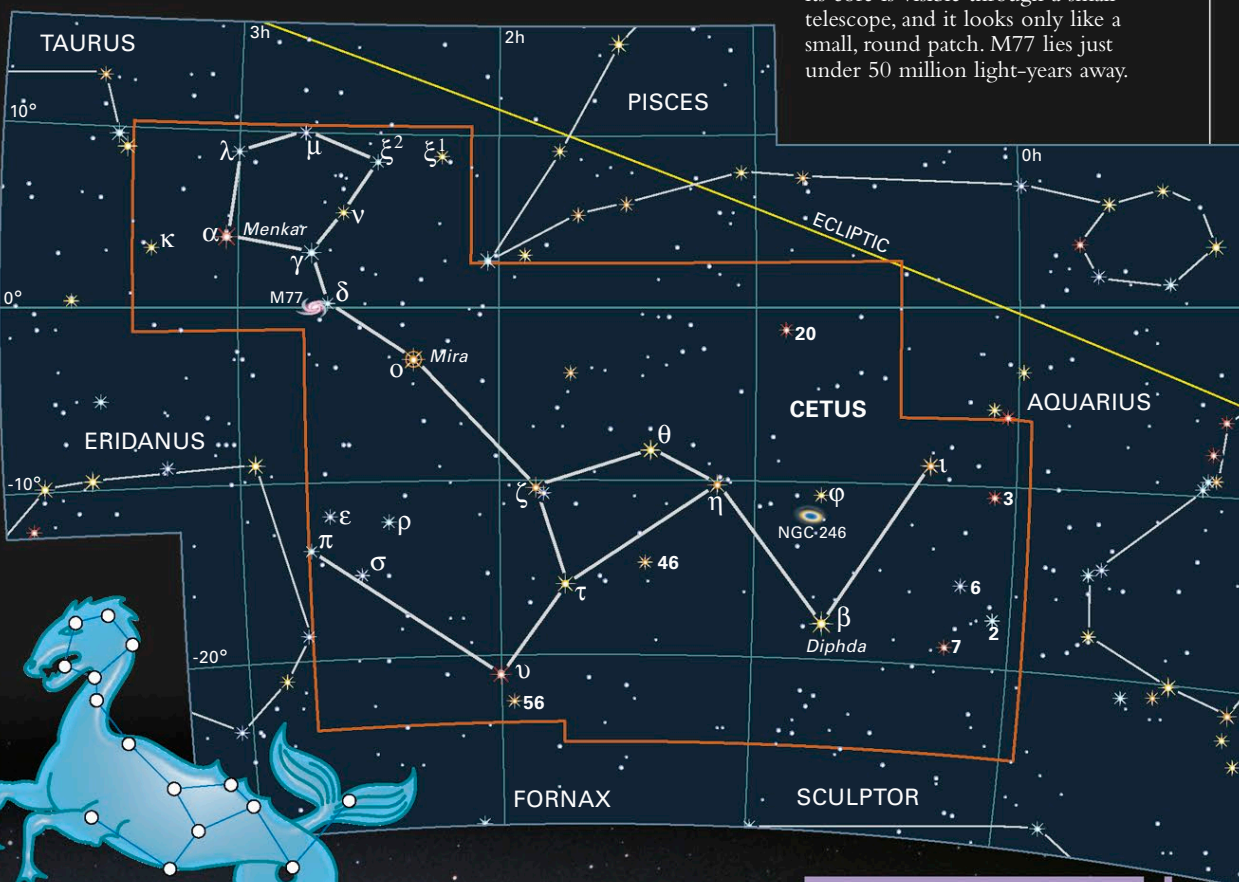
M77

Because it is a Seyfert galaxy, the spiral galaxy M77 looks like a fuzzy star through smaller telescopes—only its extremely bright core can be seen.

devastating bombardments. Thus the prospects for life in its vicinity seem rather slim.

M77 is found near Delta (δ) Ceti. This spiral galaxy is the brightest example of a Seyfert galaxy (see Types of Active Galaxies, p.320). Related to quasars, Seyfert galaxies are a class of galaxies that have extremely bright centers. M77 is oriented face-on toward the Earth, although only its core is visible through a small telescope, and it looks only like a small, round patch. M77 lies just under 50 million light-years away.

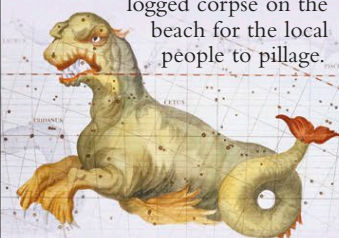
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MYTHS AND STORIES

THE SEA MONSTER

Cetus was the sea monster sent to devour the princess Andromeda in the famous Greek myth (see p.368). On his return from killing Medusa the Gorgon, Perseus spied Andromeda's plight and swooped down on the sea monster as it attacked, stabbing it repeatedly with his sword in a fury of blood and foam, and leaving its waterlogged corpse on the beach for the local people to pillage.



MYTHICAL MONSTER

Old star charts depict Cetus with enormous jaws and a coiled tail, its flippers dipped in the neighboring constellation, the river Eridanus.

LURCHING MONSTER

Cetus is large but not particularly prominent. Its most celebrated star is the variable red giant Mira (ο), which for much of the time is too faint to be seen with the naked eye.